Emile Pierret

Born on January 21, 1997.
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https://emilepi.github.io

Education

2025 – Present — **Post-doc**. *Université Paris Cité, ENS, France*
Advisor: Julie Delon.

2022 – 2025 — **Ph.D. in Mathematics**. *University of Orléans, France*
Thesis topic: *Stochastic Super-resolution and Inverse Problems: From Gaussian Conditional Sampling to Diffusion Models*.
Ph.D. advisor: Bruno Galerne.

2021 – 2022 — **Master's Degree (M2) in Mathematics, Vision, Learning (MVA)**. *ENS Paris-Saclay, Gif-sur-Yvette, France*
Research-oriented Master's program.

2020 – 2021 — **French National Agrégation in Mathematics (external track)**. *ENS Paris-Saclay, Gif-sur-Yvette, France*
Option B.

2019 – 2020 — **Master's Degree (M1) – Jacques Hadamard Program**. *ENS Paris-Saclay, Cachan, France*
Research internship: *Uncertainty quantification in COVID-19 spread: lockdown effects*.
Supervisor: Ana Carpio, *Universidad Complutense de Madrid*.

2018 – 2019 — **Double Bachelor's Degree in Mathematics and Computer Science**. *ENS Paris-Saclay, Cachan, France*

2018 — **Admitted as a normalien at ENS Paris-Saclay (4-year program)**
Admission via the Computer Science competitive entrance exam.

2015 – 2018 — **Preparatory Classes for the French Grandes Écoles**. *Lycée Descartes, Tours, France*
MPSI (Mathematics, Physics, Engineering Science), followed by MP* (Advanced Mathematics and Physics).

Publications

Preprints

1. E. Pierret, J. Hertrich, S. Hurault, and J. Delon, *Tessellations of semi-discrete flow matching*, 2026. arXiv: 2605.07513 [cs.LG]. URL: https://arxiv.org/abs/2605.07513.
2. E. Pierret, V. Tosel, J. Delon, and A. Newson, *Flow matching for applied mathematicians.* 2026.
3. É. Pierret and B. Galerne, *Exact evaluation of the accuracy of diffusion models for inverse problems with gaussian data distributions*, 2025.

Journal papers

1. É. Pierret and B. Galerne, "Stochastic super-resolution for gaussian microtextures," *SIAM Journal on Imaging Sciences*, vol. 18, no. 2, pp. 1176–1207, 2025. DOI: 10.1137/24M1657407.
2. A. Carpio and E. Pierret, "Uncertainty quantification in covid-19 spread: Lockdown effects," *Results in Physics*, vol. 35, p. 105 375, 2022, ISSN: 2211-3797. DOI: https://doi.org/10.1016/j.rinp.2022.105375.

Conference papers

1. E. Pierret and B. Galerne, “Diffusion models for gaussian distributions: Exact solutions and wasserstein errors,” in *Forty-second International Conference on Machine Learning*, 2025. URL: https://openreview.net/forum?id=bxYbxzCI2R.
2. É. Pierret and B. Galerne, “Stochastic super-resolution for gaussian textures,” in *ICASSP 2023 - 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5. DOI: 10.1109/ICASSP49357.2023.10096585.

Talks

Invited Talks

December 2026 — **From Diffusion Models to Flow Matching: A Gentle Introduction to Modern Generative Sampling.**
Journée du projet CaSciModOT, University of Orléans.

June 2025 — **A Precise Examination of Diffusion Models via Their Application to Gaussian Distributions.**
Images, Optimization and Probability Seminar, University of Bordeaux.

June 2025 — **On the accuracy of diffusion models in Bayesian image inverse problems: A Gaussian case study.**
SMAI 2025, Carcans Maubuisson.

May 2025 — **On the accuracy of diffusion models in Bayesian image inverse problems: A Gaussian case study.**
ANR MISTIC days, Lyon.

February 2025 — **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.**
Mathematical Imaging and Surface Processing Workshop, MFO Oberwolfach, Germany.

January 2025 — **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.**
Imaging in Paris Seminar, Paris.

November 2024 — **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.**
Workshop on Stochastic Geometry and Mathematics for Imaging, Nice.

October 2024 — **Stochastic super-resolution for Gaussian microtextures.**
ANR MISTIC days, Vannes.

June 2024 — **Stochastic super-resolution for Gaussian microtextures.**
LAREMA PhD Seminar, Angers.

May 2024 — **Introduction to diffusion models and their restriction to the Gaussian case.**
CANUM 2024, Île de Ré.

Talks (continued)

Posters

July 2025 — **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.**
Poster presentation, *ICML 2025*, Vancouver.

January 2025 — **Diffusion models for Gaussian distributions: Exact solutions and Wasserstein errors.**
Poster presentation, *Mathematics and Image Analysis (MIA'25)*, Paris.

June 2023 — **Stochastic super-resolution for Gaussian textures.**
Poster presentation, *ICASSP 2023*, Rhodes, Greece.

Outreach Talks and Working Group Presentations

October 2026 — **A Precise Examination of Diffusion Models via Their Application to Gaussian Distributions .**
CSD seminar, ENS Paris.

June 2024 — **Tutorial: *Introduction to Neural Networks*.**
IDP PhD Week, Courcimont Farm.

June 2023 — **Introduction to diffusion models and practical session.**
IDP Deep Learning Working Group, Orléans.

June 2023 — **Presentation of the "RePaint" method.**
Diffusion Models Working Group, MAP5, Paris.

June 2023 — **Stochastic super-resolution for Gaussian textures.**
IDP PhD Week, Courcimont Farm.

Teaching Experience

2022 – 2025 — **Lectures and practicals – Image Learning (15h)**. *University of Orléans*
Master's level (M1) in Applied Mathematics.
Introductory course on neural networks for image classification.

2023 – 2025 — **Tutorials – Algebra 3 (49h)**. *University of Orléans*
Second-year undergraduate level (L2) in Mathematics.
Tutorials in linear algebra.

2022 – 2023 — **Tutorials – Calculus (49h)**. *University of Orléans*
First-year undergraduate level (L1), first semester.
Tutorials introducing fundamental tools in analysis and algebra.

Service and Responsibilities

2024 — **Elected alternate member of the Research Committee**, University of Orléans

2022–2025 — **Organizer of the PhD seminar**, Institut Denis Poisson (Orléans–Tours)

Skills

Languages

English	**C1 level** (IELTS 7/9, 2020)
Spanish	**C1 level**

Programming

Advanced proficiency	Python, PyTorch, MATLAB, $\LaTeX$, Caml
Basic knowledge	C, C++, Assembly language, R